Data Visualization

Retail Sales Performance Analysis (2024–2025)

CodeAlpha Data Analytics Internship — Task 3

Prepared as part of the CodeAlpha Data Analytics Internship Program

Dataset: Synthetic retail transactions | 15,648 orders | Jan 2024 – Dec 2025

1. Introduction & Objective

This report fulfills Task 3 (Data Visualization) of the CodeAlpha Data Analytics internship. The goal is to transform raw retail sales data into clear visual formats that reveal trends, comparisons, and patterns useful for business decision-making.

A synthetic but realistic retail dataset was generated, simulating two years of daily transactions across six product categories and four sales regions, including seasonal effects (holiday peaks, February slowdown), weekend demand shifts, and discounting behavior.

Dataset Overview

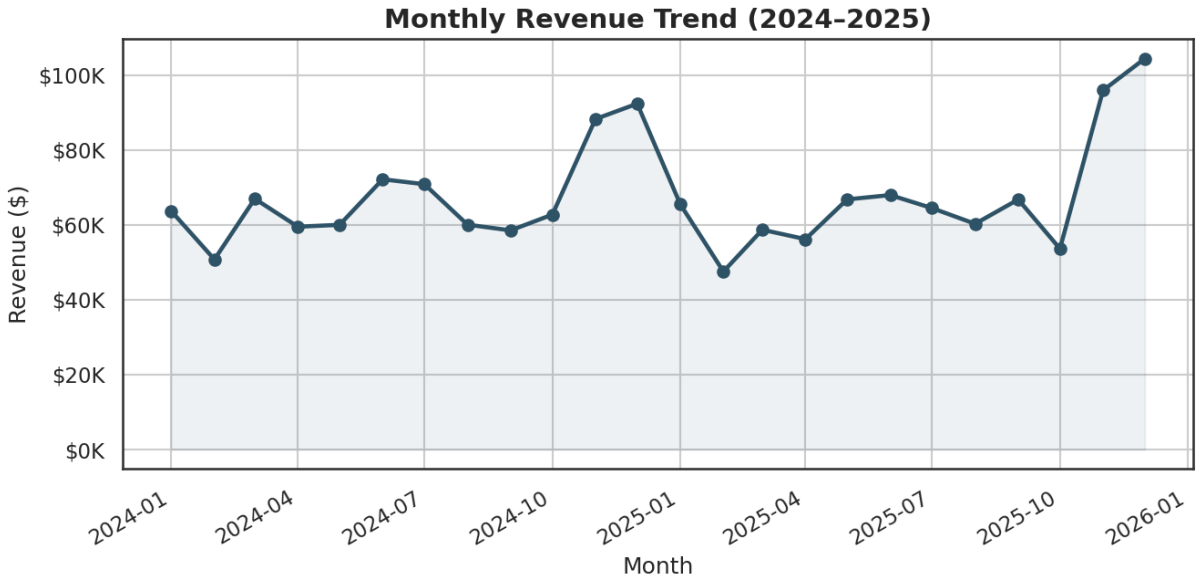
Field	Description
order_date	Date of the transaction
category	Product category (6 categories)
region	Sales region: North, South, East, West
quantity	Units purchased per order
unit_price	Price per unit ($)
discount	Discount rate applied (0–20%)
revenue	Final order revenue ($)

Total Orders	Total Revenue	Avg Order Value	Date Range
15,648	$1,613,785.84	$103.13	Jan 2024 – Dec 2025

2. Visual Analysis

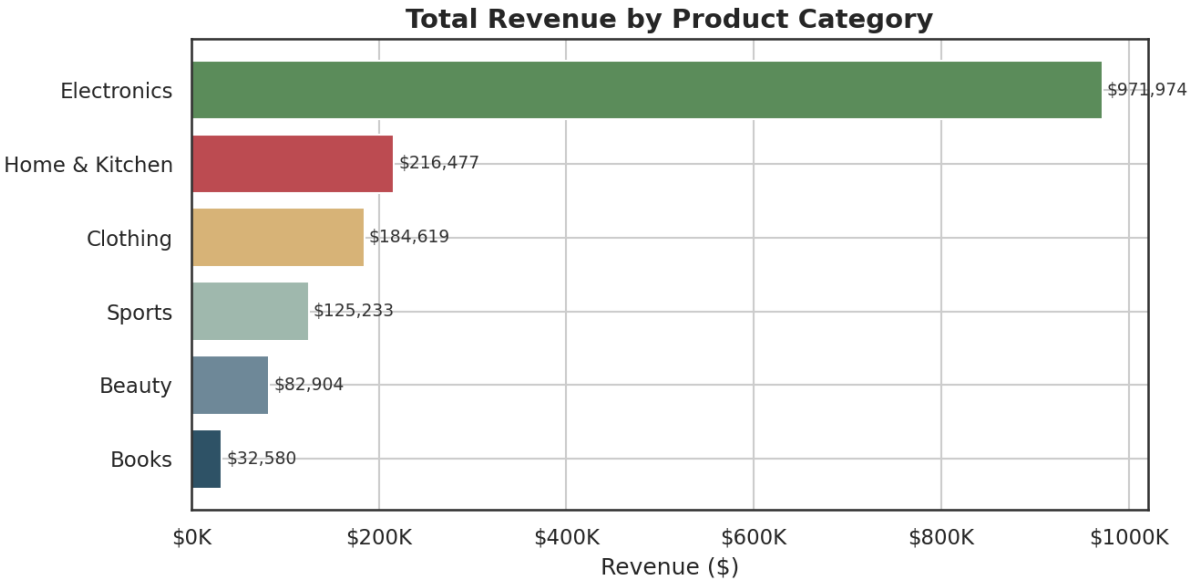
The following pages walk through six visualizations built from the retail sales dataset, each paired with a short data-driven insight.

2.1 Monthly Revenue Trend



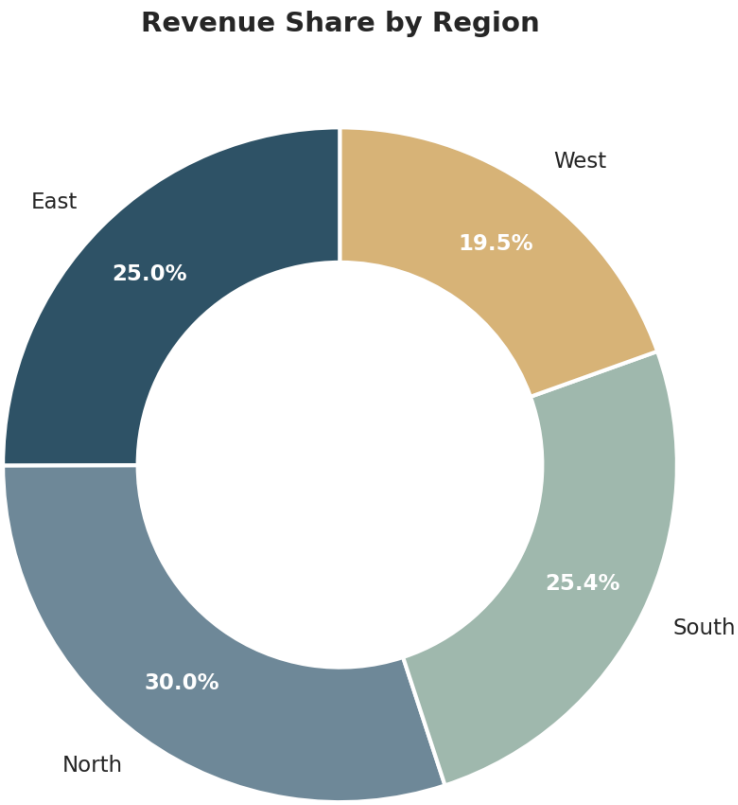
Insight: Revenue shows a strong seasonal pattern, peaking sharply in November–December (holiday shopping, ~$88K–$104K/month) and dipping in February (~$48K). This suggests inventory and marketing budgets should scale up ahead of Q4.

2.2 Revenue by Product Category



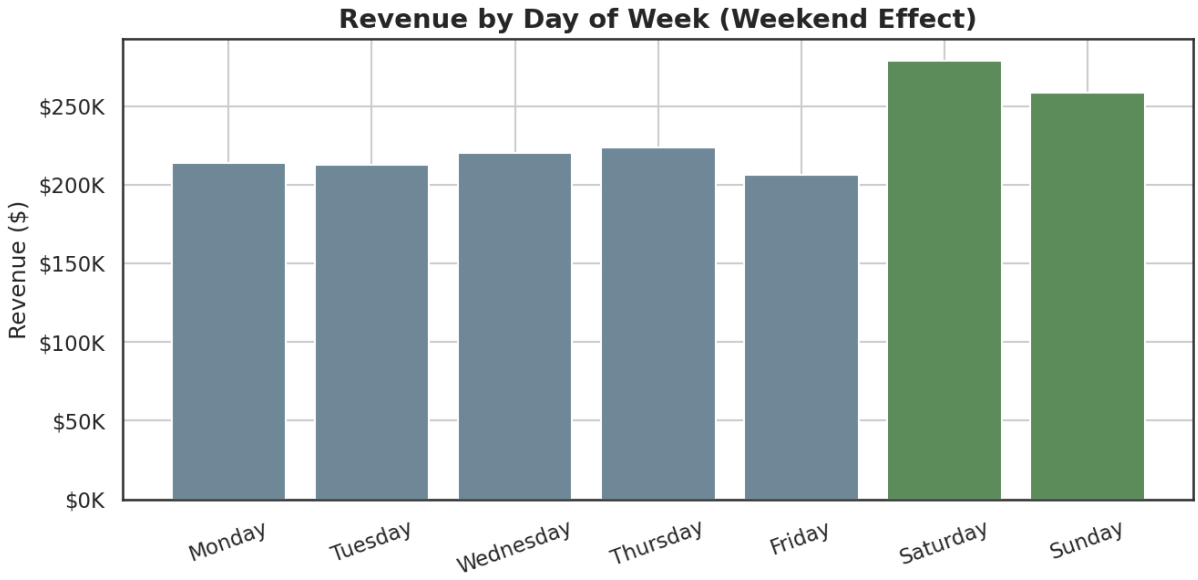
Insight: Electronics is the top-performing category at roughly $972K in total revenue, nearly 3x the next closest category, driven by its higher average unit price despite fewer units sold.

2.3 Revenue Share by Region



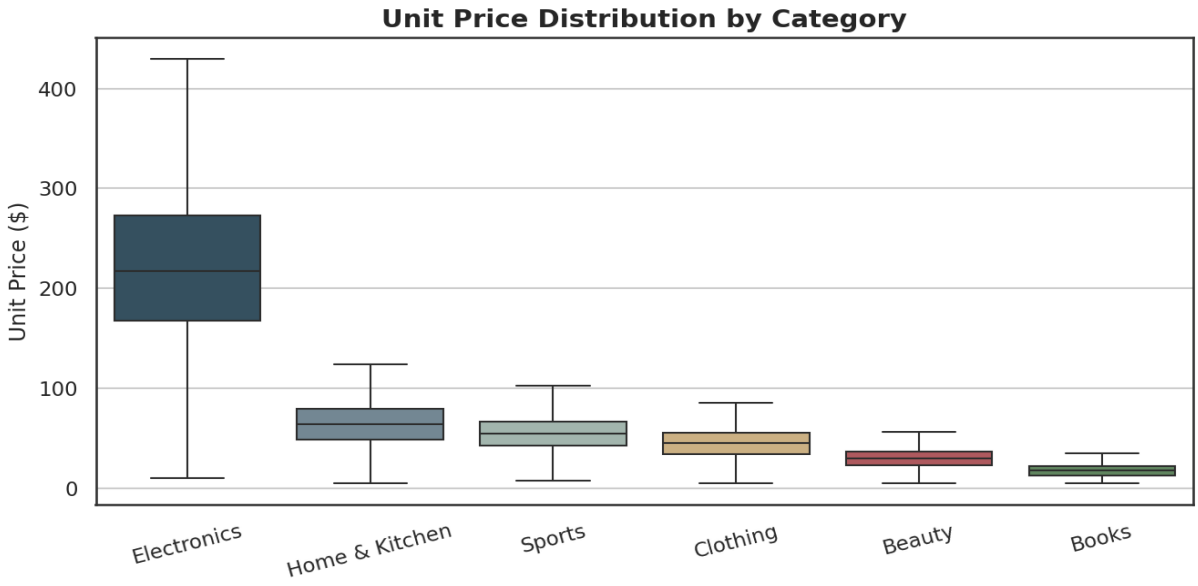
Insight: The North region leads with about 30% of total revenue, followed closely by South and East, while West lags slightly behind — a useful signal for targeted regional promotions.

2.4 Revenue by Day of Week



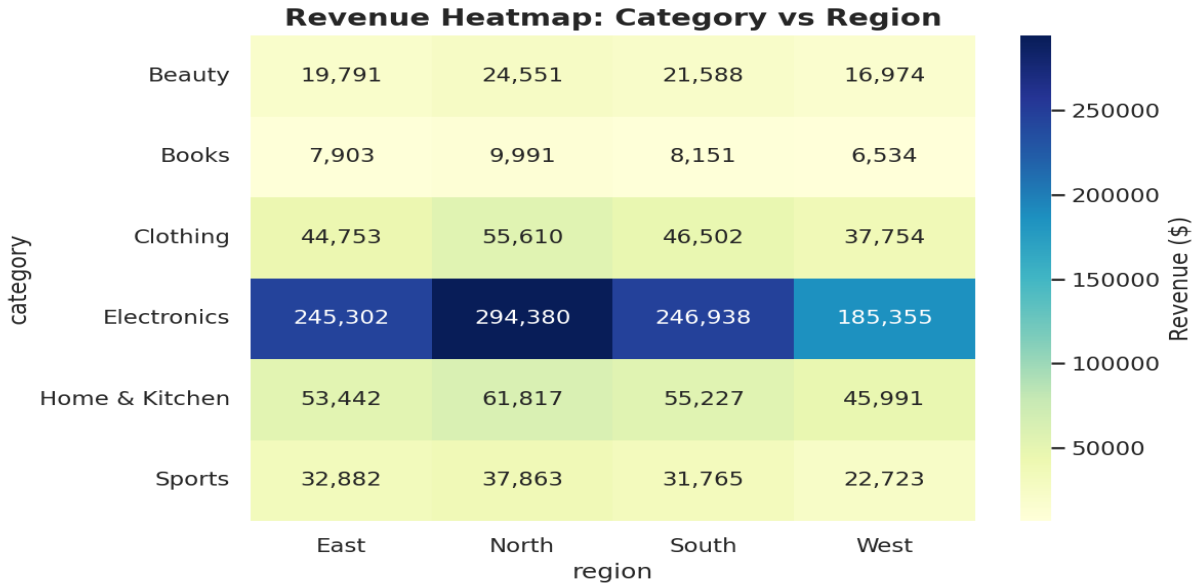
Insight: Weekend revenue (Sat/Sun) is highly competitive with weekday revenue despite fewer shopping days, confirming a per-day weekend demand boost, though weekday totals edge out slightly overall due to five days vs two.

2.5 Unit Price Distribution by Category



Insight: Electronics carries both the highest median price and widest price spread, indicating a mix of budget and premium products, while Books and Beauty are tightly clustered at lower price points.

2.6 Revenue Heatmap: Category vs Region



Insight: Electronics in the North region is the single strongest combination, suggesting regional stocking and marketing priority, while lower-revenue cells (e.g. Books in West) may indicate under-penetrated segments worth targeted campaigns.

3. Key Findings

• Seasonality is the strongest driver of revenue — Q4 (Nov–Dec) generates roughly 40–60% more monthly revenue than the yearly average.

• Electronics dominates category revenue despite representing only ~24% of orders, due to a significantly higher average order value.

• The North region is the top revenue contributor, closely followed by South and East; West presents room for growth.

• Weekend shopping shows a per-day intensity boost, useful for scheduling flash sales or ad spend.

• Discounting is applied to roughly 45% of orders, concentrated in the 10–20% range, and appears linked to higher order volume during off-peak months.

4. Recommendations

• Increase inventory and ad spend for Electronics ahead of Q4 to capitalize on peak seasonal demand.

• Launch a targeted regional campaign in the West to close the revenue gap with other regions.

• Use weekend-specific promotions to further capture the demonstrated weekend demand boost.

• Test reduced discounting in peak months, where volume is already high, to protect margins.

Tools used: Python, Pandas, Matplotlib, Seaborn — CodeAlpha Data Analytics Internship, Task 3.